Reproducibility Budgets for Open-Weight LLM Evaluation: A Generalizability Theory Analysis

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Abstract

Large Language Model (LLM) evaluation results vary across reproductions, but variance sources interact, and their relative importance shifts with the benchmark format. No unified factorial framework exists to decompose these sources simultaneously across benchmarks. We apply Generalizability theory (G-theory) in a crossed design with up to seven facets across eight open-weight 7–9B parameter models and five benchmarks, totaling approximately 2.3 million evaluation records. Our results suggest a format-associated variance pattern: on multiple-choice benchmarks, item-related variance—item difficulty plus model×item differential item functioning (DIF)—concentrates at 68–72%, while on free-form benchmarks it disperses to 31–41%; even the unbundled item component alone shows a consistent gap (34–38% vs. 17–24%), with no overlap across the five benchmarks studied. Model×item interaction (up to 36%) is the second-largest component in cross-model analysis, consistent with prior findings on differential item functioning. Under binary scoring, BF16 and FP16 numerical precisions are functionally interchangeable (<0.1% interaction). Multi-facet designs require 10× fewer items than single-condition designs to reach a generalizability coefficient of 0.80 (55 versus 552 items), and cross-model projections confirm the ratio exceeds 10× across all benchmarks, with free-form benchmarks showing nearly double the efficiency gain of multiple-choice benchmarks. Within-family scale validation (3B–9B) confirms item dominance is scale-robust. We release an open-source G-theory pipeline and D-study budget calculator for evaluation design optimization.

1 Introduction

On multiple-choice benchmarks, item selection drives 34–38% of evaluation variance while prompt wording is negligible (<3%). On the free-form reasoning task GSM8K, the structure shifts: model×prompt interaction alone accounts for approximately 9%, while item selection drops to 17%. These numbers, drawn from our factorial analysis of approximately 2.3 million evaluation records across five benchmarks, reveal that the variance structure of LLM evaluation is not a fixed property of the pipeline but shifts qualitatively with the response format. Current evaluation practice treats variance sources as independent nuisance factors and controls them one at a time: one study perturbs prompts, another varies precision, a third changes item subsets. This piecemeal approach cannot capture interactions between sources, nor can it predict how the relative importance of each source changes across evaluation settings. Leaderboard rankings built on single-condition evaluations are therefore sensitive to methodological choices that remain unreported and uncontrolled (Rodriguez et al., 2021; Dehghani et al., 2021). No unified factorial framework exists to decompose these sources simultaneously, quantify their interactions, and compare the resulting variance structure across benchmarks.

Existing studies examine individual variance sources in isolation. Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) documented 50–76% accuracy swings from prompt formatting; Yuan et al. (2025) showed up to 9% variation from numerical precision; Pezeshkpour and Hruschka (2024), Dodge et al. (2020), and Henderson et al. (2018) documented ordering and hyperparameter sensitivity—each holding other factors fixed. Messing (2026) applied G-theory to LLM evaluation across five methodology facets on a single benchmark (judge identity: 43.8%), but excluded infrastructure facets and cross-benchmark comparison. Madaan et al. (2024) proposed per-factor metrics without crossing factors. IRT approaches (Maia Polo et al., 2024; Lalor et al., 2019; Zhou et al., 2026) optimize item selection but ignore prompt, seed, and ordering facets. Bouthillier et al. (2021) established ANOVA-based variance decom-

position for ML reproducibility. Three gaps remain: no study has (a) crossed infrastructure and methodology facets in a unified factorial design, (b) compared variance structures across response formats, or (c) translated variance components into design-dependent reproducibility budgets.

We address these gaps with a multi-benchmark G-theory study (Shavelson and Webb, 1991; Brennan, 2001) crossing up to seven facets across eight open-weight 7–9B parameter models and five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, MATH), totaling approximately 2.3 million records. We define *item-related variance* as item difficulty plus model×item interaction (differential item functioning, DIF; §3.1). D-study projections show that multi-facet designs require 10× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items). We also observe a format-associated pattern: item-related variance concentrates at 68–72% on MC benchmarks and disperses to 31–41% on FF benchmarks, with no overlap across the five benchmarks studied. Model×item interaction (up to 36%) indicates that models differ primarily in *which items* they find difficult (Pezeshkpour and Hruschka, 2024; Maia Polo et al., 2024). Figure 2 visualizes the contrast.

Our contributions are as follows:

1. **Design-dependent reproducibility budgets.** Multi-facet evaluation designs require 10.0× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items), with bootstrap CIs confirming the ratio exceeds 10× across all five benchmarks. Cross-model D-study projections reveal a format-associated efficiency gap: FF benchmarks show ∼21× versus ∼12× for MC, indicating that single-condition evaluation of reasoning tasks incurs nearly double the reliability penalty.

2. **Format-associated item-related variance concentration.** In a crossed 8-model × 5-benchmark design, item-related variance (item difficulty + model×item DIF) concentrates in MC benchmarks (68–72%) and disperses in FF benchmarks (31–41%), with no overlap across the five benchmarks studied. This is reported as a descriptive observation ($n = 5$ benchmarks; permutation $p = 0.10$ at the resolution floor) with strong item-level support (Cohen's $d = 1.73$ [1.72, 1.83]). The mechanism is item difficulty heterogeneity: within MMLU's 57 subjects, difficulty dispersion correlates with item variance share ($\rho = 0.806$, $p = 0.005$).

3. **Precision separability (confirmatory).** Under binary scoring, BF16 and FP16 yield functionally identical results (<0.1% interaction with all other facets). This separability is metric-conditional: text-exact-match scoring reveals a 1.3% precision×temperature interaction absent under binary scoring.

4. **Practitioner toolkit.** We release a G-theory variance decomposition pipeline and D-study budget calculator. Practitioners can compute the number of items, models, and conditions needed for a target reliability before data collection.

2 Related Work

Variance Decomposition in ML/NLP Evaluation. Factorial decomposition of evaluation variance has roots in the ML reproducibility literature. Bouthillier et al. (2021) introduced an ANOVA-based template that decomposes benchmark variance across implementation details, hyperparameters, and data splits, establishing that design choices account for a substantial share of reproducibility failures. Several subsequent studies applied similar decomposition logic to specific facets of model evaluation (Dodge et al., 2020; Henderson et al., 2018; Magnusson et al., 2023). Messing (2026) brought Generalizability theory into LLM evaluation, decomposing variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, with prompt×judge interaction at 12.4%. However, this study used a single benchmark, included no infrastructure facets such as numerical precision, and fixed the judge as an LLM evaluator rather than isolating other sources. Madaan et al. (2024) proposed per-factor variance metrics for LLM evaluation benchmarks but did not cross factors simultaneously. Blackwell et al. (2024) quantified uncertainty in benchmark scores, and Qian et al. (2026) conducted systematic meta-evaluation of LLM benchmarks; Burnell et al. (2023) called for rethinking evaluation reporting standards in AI. Recently, G-theory has been applied to LLM-based writing assessment (Song et al., 2025) and annotation reliability (Anonymous, 2025), but these studies address scoring consis-

tency rather than benchmark evaluation variance. Kunievsky and Evans (2025) proposed an alternative decomposition that separates intent uncertainty, articulation variance, and model uncertainty, but this framework does not cross infrastructure and methodology facets in a factorial design. Our work extends these approaches by crossing seven facets across five benchmarks, enabling direct comparison of variance structures across evaluation formats.

Infrastructure and Prompt Variance. A parallel line of work examines individual variance sources in isolation. Yuan et al. (2025) documented up to 9% accuracy variation from numerical precision; Messina and Scotta (2026) and He and Thinking Machines Lab (2025) formalized and mitigated CUDA nondeterminism from floating-point non-associativity. On the methodology side, Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) found 50–76% accuracy swings from prompt formatting; Hua et al. (2025) showed that much of this sensitivity is an artifact of log-likelihood scoring (our generation-based protocol controls for this). Pezeshkpour and Hruschka (2024), Bui et al. (2025), and Biderman et al. (2024) documented ordering, seed, and training sensitivity respectively. Each study isolates one factor, precluding interaction estimation within a unified design.

IRT and Efficient Benchmarking. Item Response Theory (IRT) offers a complementary perspective on evaluation design by modeling item difficulty and discrimination parameters. Lalor et al. (2019) introduced IRT-based item analysis for NLP benchmarks, and Rodriguez et al. (2021) showed that IRT reveals hidden properties of evaluation items that raw accuracy scores obscure. Building on this foundation, Maia Polo et al. (2024) developed tinyBenchmarks, using IRT-based subsampling to reduce evaluation cost while preserving ranking accuracy, and Zhou et al. (2026) recommended a 116-item subset via PSN-IRT for efficient MMLU evaluation. Perlitz et al. (2023) and Kipnis et al. (2025) explored related strategies for benchmark compression; Li et al. (2025) extended IRT to adaptive testing. These approaches optimize item selection for latent trait estimation but do not model variance from prompt, seed, or ordering facets. IRT-selected items evaluated under a single methodological condition can therefore systematically underestimate the number of items required for reliable measurement across diverse evaluation settings.

Psychometrics Foundations. G-theory (Shavelson and Webb, 1991; Brennan, 2001) provides the canonical framework for multi-facet reliability analysis. Extensions to binary data include Mushquash and O'Connor (2006) and logistic-link generalized G-theory (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). We operationalize differential item functioning (Embretson and Reise, 2000) as model×item interaction. Recent psychometric perspectives on LLM evaluation (Kearns, 2026; Ye et al., 2025) address complementary concerns of construct validity and measurement properties.

Positioning. Relative to Messing (2026), we fix the judge (generation-based scoring) to isolate the remaining facets and extend to multiple benchmarks, revealing format-associated structure. Relative to IRT approaches (Maia Polo et al., 2024; Zhou et al., 2026), G-theory optimizes an orthogonal dimension: conditions needed for target reliability rather than item discrimination. Relative to isolated-factor studies (Yuan et al., 2025; Sclar et al., 2024; Romanou et al., 2026; Madaan et al., 2024), our factorial crossing enables interaction estimation.

3 Method

3.1 Generalizability Theory

Generalizability theory (G-theory) treats each measurement as one observation from a universe of admissible conditions defined by the crossed facets of an evaluation design (Shavelson and Webb, 1991; Brennan, 2001). The *object of measurement* is the entity whose consistency is of interest. In our setting, test items (p) serve as the object of measurement, and the goal is to determine how reliably item-level scores generalize across the methodological conditions under which they are collected.

A *G-study* decomposes total observed variance into components σ^2_x for each facet x and their interactions. Each component quantifies how much variability that source contributes to the overall measurement. Large interaction components (e.g., σ^2_{pM} for item-by-model) indicate that facets do not affect items uniformly but instead create differential difficulty profiles. We estimate variance components via Henderson Method I, an unbiased ANOVA-based estimator for balanced designs (Brennan, 2001, Ch. 3), with 1,000-resample bootstrap confidence intervals (item-level resampling with replacement, preserving the fully-crossed structure across all other facets) (Li, 2023).

A *D-study* projects the generalizability coefficient under hypothetical designs with different numbers of conditions per facet:

$$G_{\text{item}} = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_{\text{rel}}^2} \quad (1)$$

where σ_{rel}^2 aggregates all item-by-facet interaction variances, each divided by the number of conditions for the corresponding facets (Brennan, 2001, Ch. 4). This prospective capability distinguishes G-theory from classical ANOVA: rather than only attributing variance post hoc, a D-study enables optimization of evaluation designs *before* data collection (Shavelson et al., 1989). We define *item-related variance* as the sum of the item main effect (σ_p^2) and the model×item interaction (σ_{pM}^2). The latter captures differential item functioning (DIF): items whose difficulty ranking changes across models (Brennan, 2001, Ch. 4). Because DIF is operationally an item-level property—it reflects model-specific item difficulty profiles rather than a global model ability shift—aggregating $\sigma_p^2 + \sigma_{pM}^2$ provides a unified measure of item-driven variance. We note a semantic difference from educational measurement: person×item DIF reflects unidirectional person responses to items, whereas model×item DIF in LLM evaluation simultaneously reflects architecture-specific capability profiles—a dual interpretation. Bundling $\sigma_p^2 + \sigma_{pM}^2$ has psychometric justification (Brennan, 2001, Ch. 4), and we report unbundled components throughout to ensure transparency.

In ML terms, a G-study is a multi-way ANOVA where each facet is a design variable and variance components quantify each variable's contribution; a D-study projects the number of conditions per facet needed for a target reliability G, analogous to a signal-to-noise ratio where signal is item difficulty variance and noise is everything that changes across evaluation conditions.

Throughout this paper, we use the following notation: p = item, M = model, π = numerical precision, δ = prompt template, s = random seed, τ = sampling temperature, o = answer ordering, and σ_x^2 = variance component for source x.

3.2 Experimental Design: MMLU

We design a fully crossed factorial experiment (exp-001) that simultaneously varies seven facets of LLM evaluation on MMLU. Table 1 summarizes the complete design matrix, and Figure 1 illustrates the crossing structure.

Facets. Items (p): 200 MMLU questions selected via purposive stratified sampling across 10 of MMLU's 57 subjects (20 per subject, deterministic selection with seed 42). The 10 subjects span four disciplinary domains—STEM: abstract algebra, college physics, high school mathematics, computer security; humanities: world religions, philosophy; social sciences: US foreign policy, international law; professional: professional medicine, clinical knowledge—ensuring that variance estimates are not dominated by a single knowledge domain. Models (M): eight open-weight 7–9B parameter instruction-tuned models spanning eight architecture families—DeepSeek-7B, Gemma-2-9B, InternLM2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.3-Instruct, OLMo-2-7B, Qwen3-8B, and Yi-1.5-9B (full identifiers in Appendix E)—treated as a quasi-random facet (§3.6). Numerical precision (π): BF16, FP16, and FP32, all evaluated with `enforce_eager=True` for fair comparison across precision levels. Prompt template (δ): six surface perturbations of the evaluation instruction (Sclar et al., 2024), varying option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue for MC benchmarks; for FF benchmarks, templates vary chain-of-thought elicitation strategy (full template text in Appendix E). Random seed (s): six fixed values (42, 123, 456, 789, 1024, 2048). Sampling temperature (τ): three levels (0.0, 0.3, 0.7) spanning deterministic to moderately stochastic decoding. Answer ordering (o): four permutations of few-shot demonstration examples (for MMLU; structurally inert for other benchmarks, see §3.3) (Pezeshkpour and Hruschka, 2024).

Sample sizes. The cross-model BF16 balanced subset yields 460,800 records (8 models × 200 items × 6 prompts × 6 seeds × 2 temperatures × 4 orderings). A single-model analysis (Llama) spans all three precisions, yielding 259,200 binary observations across 1,296 fully balanced conditions. The BF16 subset serves as the primary cross-model sample because BF16 is the standard inference precision for these architectures.

3.3 Cross-Benchmark Extension

To test whether the variance structure observed on MMLU generalizes across evaluation formats, exp-002 extends the design to five benchmarks: MMLU, ARC-Challenge, and HellaSwag (multiple-choice), and GSM8K and MATH (free-form generation).

Table 1: Fully crossed G-theory design. Exp-001 evaluates MMLU across all facet levels; exp-002 extends to five benchmarks with reduced crossing. †BF16 balanced subset for cross-model analysis.

Facet	Levels	exp-001	exp-002
Item (p)	Stratified sample	200	200×5
Model (M)	8 architecture families	8	8
Precision (π)	BF16 / FP16 / FP32	3	1
Prompt (δ)	Paraphrases	6	6
Seed (s)	Fixed set	6	6
Temperature (τ)	0.0 / 0.3 / 0.7	3	2
Ordering (o)	Demo permutations	4	1
Total records		460,800†	2,304,800

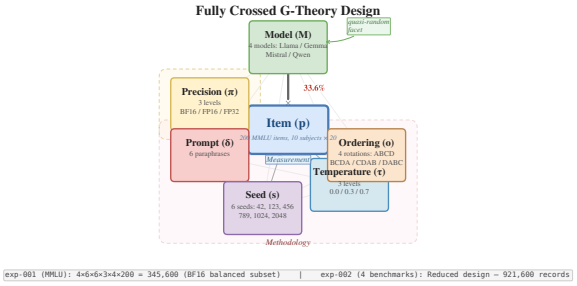


Figure 1: Fully crossed G-theory design for LLM evaluation. Each observation is a binary correctness score determined by the crossing of seven facets: item (p), model (M), precision (π), prompt (δ), seed (s), temperature (τ), and ordering (o).

The design crosses 8 models × BF16 × 2 temperatures (0.0, 0.7) × 6 prompts × 6 seeds × 4 orderings × 200 items per benchmark, totaling 2,304,000 records. This reduced crossing fixes precision at BF16 and uses two temperature levels to concentrate statistical power on format-associated differences. The ordering facet operates only on MMLU, where it selects among four permutations of the few-shot demonstration examples. For all other benchmarks, the prompt construction does not use the ordering parameter; ordering is structurally inert ($\sigma^2_{\text{ordering}} = 0$ by construction) and is retained solely for design balance. The two temperature levels bracket the most common evaluation configurations: deterministic ($\tau = 0.0$) and a moderate stochastic setting ($\tau = 0.7$) representative of practical deployment. A subset analysis confirms that this reduced crossing preserves the variance structure of the full design ($\Delta G = 0.0008$; Appendix A).

3.4 Evaluation Protocol

All evaluations use generation-based scoring (free-text response with answer extraction) rather than log-likelihood ranking, following Hua et al. (2025). Binary correctness (1/0) is the primary metric; text-exact-match is secondary. Inference uses vLLM 0.9.2 without batch-invariant kernels, preserving CUDA-level nondeterminism as a measurable variance source (Messina and Scotta, 2026; He and Thinking Machines Lab, 2025). Few-shot protocols: 5-shot MMLU/ARC, 4-shot MATH, 8-shot GSM8K, 10-shot HellaSwag.

3.5 Binary Data Assumptions

Standard G-theory assumes continuous, approximately normal outcome data (Shavelson and Webb, 1991). Our binary correctness scores (0/1) violate this assumption. We establish robustness through simulation and estimator properties, with two additional analyses (GLMM complement, accuracy-range stability) in Appendix A.

A probit simulation matched to our 1,296-condition design confirms that Henderson Method I recovers variance component rankings from dichotomized data (Spearman $\rho = 0.732$ between binary and continuous rankings; `item_id` dominant in all 100 replications). Henderson Method I produces unbiased estimates for balanced designs regardless of outcome distribution (Brennan, 2001, Ch. 12), and REML estimates differ by <0.5pp on all components (Appendix A).

Our target universe comprises open-weight, instruction-tuned models in the 7–9B parameter range, sampled from six architecture families. Results may not generalize to API-based models (where infrastructure is opaque), base (non-instruction-tuned) models, or models above 9B parameters. Treating eight purposively selected models as a quasi-random facet within this universe is validated by two analyses (Appendix A): four-model subset D-study projections predict $G = 0.893$ at eight models (observed: $G = 0.896$, $\Delta = 0.003$), and all eight leave-one-model-out subsets preserve component rankings.

4 Experiments

All experiments use open-weight 7–9B parameter models evaluated with vLLM 0.9.2 on a single NVIDIA RTX PRO 6000 GPU (96GB) with binary correctness scoring.

4.1 Single-Model Baseline

We first establish the variance structure within a single model (Llama-3.1-8B-Instruct) on MMLU

to isolate methodological facet contributions before introducing cross-model variation.

Design. The G-study crosses six facets: 200 items × 3 numerical precisions (BF16, FP16, FP32) × 6 prompt templates × 6 random seeds × 3 temperature levels (0.0, 0.3, 0.7) × 4 answer orderings, yielding 259,200 binary observations. Variance components are estimated via Henderson Method I with bootstrap confidence intervals (1,000 resamples).

Results. Under binary scoring, item difficulty (`item_id`) accounts for 58.4% of total variance, with the residual term contributing 23.1%. All main facet effects are negligible (<1% each): precision 0.9%, prompt template, temperature, seed, and ordering each <0.5%. The interesting structure appears in item-conditional interactions: prompt×item is the largest at 5.5%, followed by seed×item (4.7%), precision×item (3.3%), temperature×item (2.9%), and ordering×item (0.8%). This pattern indicates that methodological choices do not shift performance uniformly across items; instead, specific items are differentially sensitive to specific facets—prompt template variation produces the largest item-conditional effect.

The resulting generalizability coefficient is $G_{\text{item}} = 0.936$, indicating that 200 items under multi-facet conditions produce highly reliable rankings. A D-study projection shows $G \geq 0.80$ is achievable with as few as 55 items in this multi-facet design.

For text-exact-match scoring (requiring verbatim string matches), the variance structure shifts dramatically: temperature dominates at 79.7% of variance, and G_{item} collapses to 0.007. This metric-conditional divergence motivates our exclusive use of binary correctness for the remaining analyses.

4.2 Cross-Model Variance

Extending to eight models (BF16 subset, 460,800 records), Table 2 shows that item difficulty drops from 58.4% to 37.9%, while model×item interaction emerges at up to 36%. The model main effect is 3–5% (MC) and 9–11% (FF), confirming that models differ primarily in *which items* they find difficult rather than in overall ability level.

The per-model interaction magnitudes reveal meaningful asymmetry, suggesting that instruction-tuning strategies produce model-specific item difficulty profiles.

Table 2: Variance decomposition for the cross-model MMLU design (8 models, BF16, 460,800 records). Components estimated via Henderson Method I. $G_{\text{item}} = 0.896$.

Component	Var.%
item_id	**37.9**
model×item	30.2
residual	25.0
model	2.8
prompt×item	2.5
model×prompt	1.6
all others	<1.0

4.3 Format-Associated Variance Structure

We extend to five benchmarks (three MC: MMLU, ARC-Challenge, HellaSwag; two FF: GSM8K, MATH) using exp-002 (8 models × BF16 × 2 temperatures × 6 prompts × 6 seeds × 4 orderings × 200 items; MATH uses 7 models due to incomplete DeepSeek inference data). Table 3 reveals a format-associated pattern. For all three MC benchmarks, item-related variance (item + model×item DIF; e.g., MMLU: 37.9% + 30.2% = 68.1%) concentrates at 68–72%. HellaSwag is the sole MC case where model×item slightly exceeds item (36.0% vs. 34.3%), though their sum remains in the MC range.

Free-form benchmarks differ markedly. On GSM8K, variance distributes across item (16.9%), model×item (14.6%), model (10.7%), and model×prompt (8.5%); prompt main effect is 5.4%. MATH shows an intermediate pattern with 7 models (DeepSeek excluded): item leads at 24.0% with model×item at 17.3%, but residual variance is substantially larger (43.9%). Cross-benchmark LOMO analysis shows that removing any single model changes G_{item} by less than 0.02 across all four benchmarks with complete 8-model data (Appendix A). The 7-model MATH analysis may conservatively underestimate model-related variance, as DeepSeek exclusion shifts variance from model×item to residual based on these cross-benchmark LOMO trends. The large FF residuals (40–44%) likely reflect chain-of-thought path diversity and answer extraction variability. Since scoring granularity fundamentally reshapes variance structure (binary $G = 0.936$ vs. text-exact-match $G = 0.007$; §4.1), our format-associated findings are conditional on binary scoring.

The observed variance structure is format-associated: MC benchmarks concentrate item-related variance at 68–72%, while FF benchmarks

Table 3: Format-associated variance structure across five benchmarks (exp-002, 8 models; MATH: 7). Variance components (%) per benchmark; MC = multiple-choice, FF = free-form. Bold indicates the dominant non-residual component per benchmark. For MC benchmarks, "prompt" reports prompt×item (prompt main effect <1%); for FF benchmarks, it reports the prompt main effect.

	MC			FF	
	MMLU	ARC	Hella.	GSM	MATH
item	**37.9**	**37.9**	34.3	16.9	24.0
mod.×item	30.2	33.7	**36.0**	14.6	17.3
model	2.8	3.1	5.4	10.7	8.5
prompt	2.5	0.4	2.3	5.4	1.0
mod.×prmpt	1.6	8.0	0.6	8.5	2.0
residual	25.0	15.4	21.4	39.9	43.9
all others	<1.0	1.5	<1.0	4.0	3.3

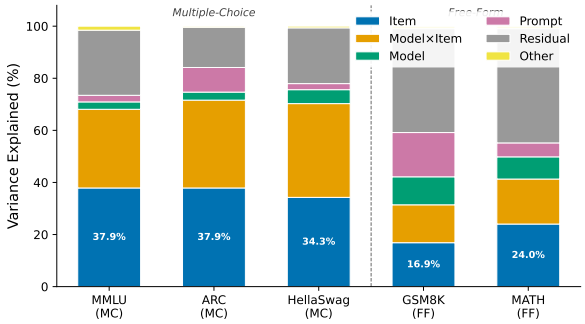


Figure 2: Variance decomposition (Henderson Method I) across five benchmarks and eight models (MATH: seven). Multiple-choice benchmarks share an item-dominated structure; HellaSwag is the exception where model×item (36.0%) exceeds item difficulty (34.3%). Free-form benchmarks (GSM8K, MATH) distribute variance across item, model, and interaction components with no single dominant source.

disperse it to 31–41%, with no overlap. Critically, even restricting to item difficulty alone (excluding model×item DIF), MC benchmarks show 34–38% versus FF 17–24%, maintaining complete separation (unbundled benchmark-level $d = 4.84$, bootstrap 95% CI $[2.32, 8.19]$), confirming that the format gap is not an artifact of DIF bundling. A permutation test yields $p = 0.10$ ($\binom{5}{3} = 10$ exhaustive partitions), the resolution floor—rejecting at $\alpha = 0.05$ is impossible regardless of effect magnitude. The item-level effect size is tightly bounded ($d = 1.73$; 95% CI $[1.72, 1.83]$), but the benchmark-level CI spans zero ($[-1.7, 5.1]$, $df = 3$). We therefore present the format-associated pattern as a descriptive observation with strong item-level support rather than a confirmatory finding. This does not extend to item budgets: GSM8K requires 128 items for $G \geq 0.80$ while HellaSwag (MC, 118) exceeds both ARC (98) and MATH (101), showing that format alone does not predict evaluation cost. Figure 2 visualizes the contrast.

One might argue the format gap is tautological: MC constrains generation noise, mechanically inflating item-related shares. Two observations argue against this. First, HellaSwag is MC yet model×item (36.0%) exceeds item (34.3%)—pure format effects would produce uniform item dominance. Second, within MMLU's 57 subjects, difficulty dispersion correlates with item variance ($\rho = 0.806$, $p = 0.005$, $n = 10$), and `high_school_mathematics` (open-ended numerical items) shows item variance of just 10.4%, below GSM8K (16.9%). Cross-benchmark std(p_i) shows a strong positive association with `item_id` share ($\rho = 0.82$, $n = 5$; model subsets vary slightly by benchmark due to data availability; Appendix D), with MC benchmarks showing higher dispersion (0.27–0.31) than FF (0.22–0.23). Format constrains the noise floor, but item difficulty heterogeneity determines the magnitude of item-related variance within that floor.

4.4 Precision Separability

Under binary scoring, BF16 and FP16 are functionally interchangeable: mean accuracy differs by 0.23 percentage points, and all precision×methodology interactions contribute <0.1% of variance (Yuan et al., 2025). Under text-exact-match scoring, a precision×temperature interaction emerges at 1.3%, consistent with the metric-conditional pattern in §4.1. For standard binary evaluation at BF16/FP16 on single-GPU inference, numerical precision is a negligible variance source that need not be controlled (Shanmugavelu et al., 2024). For non-binary metrics (e.g., BLEU, exact string match), our budgets do not apply; the text-exact-match collapse ($G = 0.007$) suggests such metrics require fundamentally larger budgets.

4.5 Design-Dependent Reproducibility Budgets

How many items does a given evaluation design need to achieve a target reliability? We translate variance components into item budgets via D-study projections.

Table 4 compares evaluation designs at the $G \geq 0.80$ threshold using single-model (Llama-3.1-8B-Instruct) MMLU variance components. The multi-facet design (1,296 conditions per item) achieves $G = 0.936$ at 200 items and requires only 55 items

Table 4: D-study reproducibility budgets under different evaluation designs (single-model, MMLU). Total evals = items × conditions.

Design	Cond.	$n_{G\geq.80}$	Total	G_{200}	G_{116}
Multi-facet	1,296	**55**	71,280	**.936**	—
Single-cond.	1	552	552	.592	.457
IRT+multi	1,296	—	150,336	—	.895‡
IRT baseline	—	116	116	—	.457†

† PSN-IRT 116 items (Zhou et al., 2026) under single-condition: $G = 0.457$.
‡ Same 116 items under multi-facet: $G = 0.895$.

for $G \geq 0.80$. A single-condition design reaches only $G = 0.592$ at 200 items and requires 552 items—a 10.0× budget ratio stable across subset sizes (CV = 1.9%).

Zhou et al. (2026)'s IRT-selected 116 items yield $G = 0.457$ under single-condition but $G = 0.895$ under multi-facet conditions, demonstrating that IRT item selection and G-theory protocol optimization are orthogonal and complementary. The higher total evaluation count (71,280 vs. 552) reflects the cost of reliable measurement. These projections have out-of-sample support: four-model subset D-study predictions match the observed eight-model coefficient within $\Delta = 0.003$ (§3.5). Budgets are validated within our model universe (open-weight, 7–9B, instruction-tuned); practitioners should verify applicability to their specific model population.

For practitioners seeking a minimal upgrade, 3 prompts × 3 seeds ($G_{200} = 0.766$, $n_{G\geq0.80} \approx 140$) is the cheapest design exceeding $G = 0.80$; a full adoption gradient is in Appendix C. Cross-model D-study projections (Appendix B) reveal that the efficiency gap is itself format-associated: MC benchmarks show 11–13× ratios while FF benchmarks show 21–22× (bootstrap CIs confirm all exceed 10×), making multi-facet designs especially critical for reasoning tasks. Figure 3 plots the D-study curves; the multi-facet curve saturates beyond ∼150 items, while the single-condition curve does not reach $G = 0.80$ within typical benchmark sizes.

4.6 Preliminary Scale Validation

A within-family analysis (Qwen2.5 3B and 7B on MMLU, same design) confirms item dominance is scale-robust: `item_id` accounts for 75.3% (3B) and 73.4% (7B), with $G = 0.963$ and 0.956 respectively. Minor scale-dependent prompt sensitivity appears (prompt×item: 12.2% at 7B vs. 8.8% at 3B). Extension to 70B+ models remains for future work.

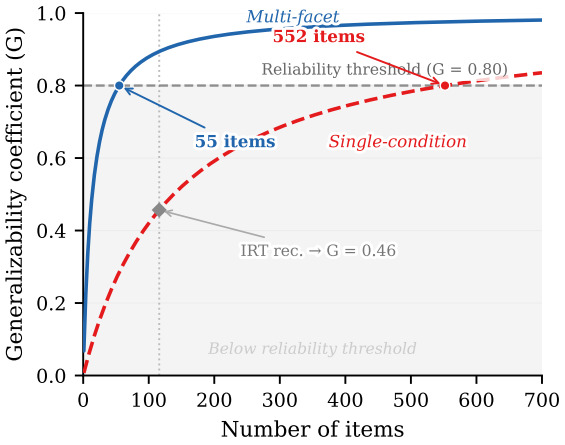


Figure 3: D-study projection curves (single-model, MMLU). The multi-facet design (10.0× fewer items) reaches $G \geq 0.80$ at 55 items; the single-condition design requires 552 items. Cross-model per-benchmark projections are in Appendix B. Shaded band: 95% bootstrap CIs.

4.7 Robustness Checks

Three independent 200-item MMLU samples yield item difficulty variance shares of 73.2%, 72.6%, and 69.0% (CV = 3.1%), confirming sampling stability. Seven additional analyses (GLMM, accuracy-range, estimator comparison, CI method, facet specification, LOMO, null calibration) in Appendix A all confirm stability.

5 Conclusion

Multi-facet designs require 10× fewer items than single-condition designs for $G \geq 0.80$ (55 vs. 552), confirmed across all five benchmarks (bootstrap CIs exceed 10×); FF benchmarks show nearly double the gain (∼21× vs. ∼12× for MC). We also observe a format-associated variance pattern: item-related variance concentrates at 68–72% on MC benchmarks and disperses to 31–41% on FF benchmarks—a descriptive observation whose confirmatory generalization requires additional benchmarks. BF16 and FP16 are interchangeable under binary scoring. Code, data, and D-study budget calculator will be released upon publication.

6 Limitations

Our findings are bounded by four scope constraints. First, the primary cross-model analysis covers eight open-weight 7–9B parameter instruction-tuned models; within-family scale validation extends to 3B–9B (Qwen2.5), but results do not generalize to API-based models (where infrastructure

is opaque), base (non-instruction-tuned) models, models above 9B parameters, or domain-adapted fine-tuned variants. Extending this validation to larger scales (e.g., 70B+) remains for future work; larger models with higher accuracy could exhibit compressed item variance due to ceiling effects. Second, all experiments run on single-GPU inference (NVIDIA RTX PRO 6000); multi-GPU tensor parallelism introduces additional numerical non-associativity that our design does not capture. Third, the benchmark sample covers three multiple-choice and two free-form tasks ($n = 5$); item-level variance patterns within each benchmark are well-powered, but the format-associated claim rests on a benchmark-level comparison with limited statistical power ($p = 0.10$, permutation test). Other evaluation formats such as code generation, long-form text production, and multi-turn dialogue may exhibit different variance structures. Fourth, all variance decompositions use binary (correct/incorrect) scoring. Partial-credit or rubric-based scoring could alter free-form variance structure—in particular, capturing chain-of-thought quality may redistribute residual variance (currently 40–44% for FF) to item or model$\times$item components, potentially narrowing the format-associated gap. Two natural extensions remain: multi-GPU tensor parallelism (testing precision separability under distributed numerical non-associativity) and fine-tuning variance analysis (quantifying how adaptation reshapes item difficulty profiles).

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A Additional Robustness Analyses

We report eight robustness analyses that verify the stability of the variance decomposition under alternative analytical choices.

GLMM complement. A generalized linear mixed model (GLMM) with marginal logit link produces variance component rankings that perfectly preserve all seven grouped component orderings ($\rho = 1.0$) relative to Henderson Method I (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). This confirms that the linear variance decomposition is robust to the binary outcome distribution.

Accuracy-range stability. The `item_id` variance share remains stable across subsets with mean accuracy from 0.62 to 0.72 ($r = -0.048$). This rules out distortion from binary floor or ceiling effects at observed accuracy levels.

Estimator robustness. We compare Henderson Method I with restricted maximum likelihood (REML-EM, 500 iterations, practical convergence $\Delta < 0.003$ percentage points). All variance components differ by <0.5 percentage points between estimators. The largest discrepancy occurs on the residual component (0.47pp); all substantive components (item, model×item, model) differ by <0.3pp. Henderson Method I is preferred for our balanced designs because it produces unbiased estimates without distributional assumptions (Brennan, 2001, Ch. 12).

CI method comparison. Bootstrap (1,000 item-level resamples with replacement) and delete-1 jackknife confidence intervals agree closely: width ratios range from 0.96 to 1.04 across all variance components. The agreement indicates that the dependence structure introduced by item-level bootstrap resampling (which creates ∼127 unique items per 200-item resample) does not materially distort interval estimates. Stratified resampling by subject (10 subjects × 20 items) produces CIs within 4% of simple item-level bootstrap, confirming that the within-subject dependence structure does not materially affect interval estimates.

Facet specification sensitivity. Treating temperature and ordering as fixed (rather than random) facets changes G_{item} by only 0.37 percentage points (all-random: 0.820 vs. fixed temperature+ordering: 0.824). The small difference reflects the negligible temperature×item (0.65%) and ordering×item (0.54%) variance shares. The random-facet assumption is therefore robust to facet classification.

Leave-one-model-out (LOMO). All eight seven-model subsets preserve the item + model×item dominance pattern: item_id ranges from 36% to 41%, and G_{item} ranges from 0.877 to 0.889. No single model disproportionately drives the observed variance structure. The consistency of component rankings across all subsets supports the quasi-random model facet assumption (§3.5). Table 5 extends the LOMO analysis to all four benchmarks with complete 8-model data, confirming that $|\Delta G_{\text{item}}| < 0.02$ uniformly.

Table 5: Cross-benchmark DeepSeek LOMO: effect of removing DeepSeek from each benchmark with complete 8-model data.

Benchmark	Format	ΔG_{item}	Δitem%
ARC	MC	+0.002	+3.5
HellaSwag	MC	−0.006	+2.6
GSM8K	FF	−0.009	+0.4
MMLU	MC	−0.018	+1.3

Null calibration. We generate 100 Bernoulli-distributed datasets with item difficulties matched to the empirical distribution but no systematic facet effects. The observed-to-null variance ratio is 1.0008, confirming that Henderson Method I does not introduce systematic bias when applied to binary outcomes. The null simulation also verifies that the item variance share (58.4%) is not an artifact of the Bernoulli $p(1-p)$ ceiling: the null produces item variance of 0.08%, three orders of magnitude below the observed value. Across all 100 replications, the same three components (item, model×item, residual) emerge as the top three (set consistency 100%), with Kendall's $\tau = 0.81$ between binary and continuous rankings.

Reduced crossing validation. To verify that the reduced crossing in exp-002 (BF16 only × 2 temperatures) does not introduce confounding, we extracted the corresponding subset from the fully crossed exp-001 design. The generalizability coefficient differs by only 0.0008 ($G = 0.9361$ vs. 0.9369), and all non-precision variance components differ by less than 2 percentage points. The 10pp increase in item variance share (58.4% → 68.5%) is fully explained by the removal of precision-related components (∼4.5% total), which redistributes across remaining sources. This confirms that the exp-002 design preserves the variance structure of the full crossing.

B Cross-Model D-Study Projections

Table 6 extends the single-model D-study (Table 4 in main text) to per-benchmark cross-model projections with bootstrap confidence intervals (1,000

item-level resamples, 8 models, 2,304 conditions per item). The efficiency ratio is format-associated: MC benchmarks show 11–13× (mean 12.0× [11.6, 12.5]), while FF benchmarks show 21–22× (mean 21.5× [20.4, 22.8]).

Table 6: Cross-model D-study reproducibility budgets with 95% bootstrap CIs. $n_{G\geq 0.80}$: items needed. Multi = multi-facet (2,304 cond.), Single = single-condition.

	Multi $n_{G\geq 0.80}$		Single $n_{G\geq 0.80}$		Ratio
	Point	95% CI	Point	95% CI	[95% CI]
Multiple-choice					
MMLU	93	[76, 116]	1216	[1004, 1491]	13.1 [12.5, 13.7]
ARC	98	[71, 139]	1078	[793, 1512]	11.0 [10.7, 11.4]
Hella.	118	[93, 151]	1403	[1120, 1790]	11.9 [11.5, 12.3]
Free-form					
GSM8K	128	[103, 162]	2768	[2230, 3531]	21.6 [20.7, 23.0]
MATH	101	[86, 122]	2160	[1853, 2631]	21.4 [20.2, 22.5]

The higher item requirements relative to the single-model analysis (55 items in Table 4) reflect the additional model×item interaction variance that enters σ^2_{rel} when models are crossed as a facet. The format-associated ratio pattern—FF benchmarks showing ∼2× the MC ratio—is consistent with the distributed variance structure observed in §4.3, where FF benchmarks allocate more variance to model and model×prompt components. MATH results use verified per-model inference across 7 models; DeepSeek was excluded due to incomplete inference data (37,400/57,600 expected records).

C Intermediate D-Study Designs

Table 7 shows how reliability improves as evaluation designs increase from single-condition to full multi-facet crossing. The steepest gain occurs at the "easy upgrade" level (2 prompts × 2 seeds = 4 conditions): G_{200} jumps from 0.592 to 0.692 at 4× cost. A minimal sufficient design for $G \geq 0.80$ is 3 prompts × 3 seeds (9 conditions, $G_{200} = 0.766$, $n_{G\geq 0.80} \approx 140$). Beyond the "standard" level (32 conditions), adding orderings provides negligible benefit (ordering×item <1%), confirming that prompt and seed diversity are the primary reliability drivers.

D Cross-Benchmark Difficulty Heterogeneity

Figure 4 plots item difficulty heterogeneity (std(p_i) across models) against `item_id` variance share for all five benchmarks. MC benchmarks cluster at higher dispersion (0.27–0.31) and higher item variance (34–38%), while FF benchmarks occupy the lower-left region (0.22–0.23, 17–24%). The cross-benchmark Spearman $\rho = 0.82$ ($p = 0.089$, $n = 5$) is consistent in direction with the within-MMLU correlation ($\rho = 0.806$, $p = 0.005$, $n = 10$ subjects), providing multi-scale support for the item difficulty heterogeneity mechanism discussed in §4.3.

Table 7: Intermediate D-study designs (single-model, MMLU). Designs are ordered by increasing conditions per item; G_{200} and $n_{G\geq 0.80}$ are D-study projections from exp-001 variance components.

Design	Conditions	G_{200}	$n_{G\geq .80}$
Single-condition	1	.592	552
Easy upgrade (2p×2s)	4	.692	∼245
Moderate (3p×3s)	9	.766	∼140
Recommended (3p×3s×2t)	18	.818	∼95
Standard (4p×4s×2t)	32	.869	∼75
Enhanced (+4 orderings)	72	.871	∼73
Full multi-facet	1,296	.936	55

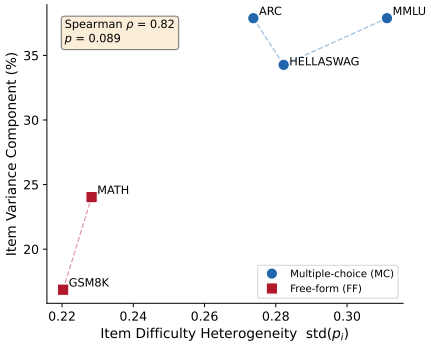


Figure 4: Item difficulty heterogeneity (std(p_i), computed over available models per benchmark, 6–8 models) versus `item_id` variance share (as reported in Table 3) across five benchmarks. MC benchmarks (filled) exhibit higher difficulty dispersion and item variance than FF benchmarks (open). Spearman $\rho = 0.82$, $p = 0.089$.

E Model and Prompt Details

Models. Table 8 lists the eight models with their HuggingFace identifiers.

Table 8: Models used in all experiments.

Short name	HuggingFace identifier	Params
Llama-3.1-8B	`meta-llama/Llama-3.1-8B-Instruct`	8B
Gemma-2-9B	`google/gemma-2-9b-it`	9B
Mistral-7B-v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	7B
Qwen3-8B	`Qwen/Qwen3-8B`	8B
Yi-1.5-9B	`01-ai/Yi-1.5-9B-Chat`	9B
OLMo-2-7B	`allenai/OLMo-2-1124-7B-Instruct`	7B
InternLM2.5-7B	`internlm/internlm2_5-7b-chat`	7B
DeepSeek-7B	`deepseek-ai/deepseek-llm-7b-chat`	7B

Prompt templates. Our six prompt templates follow a surface perturbation design inspired by BrittleBench (Romanou et al., 2026). For multiple-choice benchmarks, templates vary along three dimensions: option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue. For free-form benchmarks, templates vary chain-of-thought elicitation strategy. All six templates are semantically equivalent—they request the same task with different surface forms—to isolate formatting effects from content effects. ARC-Challenge and HellaSwag use the same six surface patterns as MMLU with domain-appropriate system instructions; MATH mirrors the GSM8K pattern with `\boxed{}` answer formatting.

MMLU templates (representative MC format):

1. `The following are multiple choice questions (with answers) about {subject}.\n\n{question}\nA. ... D. ...\nAnswer:`
2. `Answer the following {subject} questions by selecting the correct option.\nQuestion: {question}\n(A) ... (B) ... (C) ... (D) ...\nCorrect answer:`
3. `Below are {subject} multiple-choice problems. Pick the right letter.\nQ: {question}\n A) ... B) ... C) ... D) ...\nA:`
4. `Answer the following {subject} question by selecting A, B, C, or D.\n{question}\n- A. ... - B. ... - C. ... - D. ...\nAnswer:`
5. `Below is a {subject} exam question with four possible answers. Choose the correct one.\n{question}\nOption A: ... Option B: ... Option C: ... Option D: ...\nCorrect:`
6. `Question ({subject}):\n{question}\nChoices: (A) ... (B) ... (C) ... (D) ...\nYour answer:`

GSM8K templates (representative FF format):

1. `Solve the following math problem step by step.\nQ: {question}\nA: Let's think step by step.`
2. `Answer the following math word problem. Show your work.\nQuestion: {question}\nSolution:`
3. `Solve each problem below. Write your reasoning, then give the final answer after ####.\nProblem: {question}\nReasoning:`
4. `You are a math tutor. Solve the problem and show your steps.\n{question}\nSteps:`
5. `Work through this math question carefully.\nQ: {question}\nWork:`
6. `Math problem:\n{question}\nLet's solve this step by step.`